

On the Effectiveness of Conveying BIM Metadata in VR Design Reviews for Healthcare Architecture

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ABSTRACT

This research seeks to assess whether Virtual Reality (VR) can be used to convey Building Information Modelling (BIM) metadata alongside geometric and spatial data in a virtual environment, and by doing so, determine if this increases the understanding of the design by the stakeholder. A user study assessed participants performance and preference for conducting design reviews in VR or using a traditional design review system of PDF drawings and a 3D model. Early results indicate VR was preferred with fewer errors made during assessment and a higher System Usability Scale (SUS) score.

Keywords: BIM, Virtual Reality, Architecture.

Index Terms: Human-centered computing – Human computer interaction (HCI) – Interaction paradigms – Virtual reality. Applied computing – Arts and humanities – Architecture (buildings) – Computer-aided design. General and reference – Cross-computing tools and techniques – Design

1 INTRODUCTION

Typically Architects and other architecture, engineering and construction (AEC) consultants communicate their designs to stakeholders via plans, sections and elevations [1]. This mode of communication has proven to be problematic as people outside the AEC industry have difficulties understanding these 2D drawings [2]. This is troublesome in healthcare projects where clients have specialised requirements that need to be accurately defined to the design team. [3].

Design Reviews in VR have shown to be effective in communicating spatial design intent to stakeholders [4]. However, the success of a healthcare project in particular depends on more than spatial configurations. Specialised equipment with specific requirements need to be correctly coordinated and allowed for [5]. The specifications and particulars of this equipment is stored in product data sheets or schedules that are separate from the plans and sections. Within a BIM project this information is coded into each object in the form of BIM metadata. This paper seeks to determine if VR can be used to effectively convey this BIM metadata to aid stakeholders in understanding the complexities of a design, in particular the requirements of specialised equipment.

To test if viewing designs and BIM metadata in VR is beneficial to the design review process, a user study was conducted during which healthcare professionals reviewed the design of a hospital in 2 different conditions - PDF drawings alongside a 3D model and a VR simulation. During the reviews the participants completed a series of tasks designed to measure their understanding of the design. Afterward they completed an SUS questionnaire and a short interview regarding the tasks just completed.

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2 BACKGROUND

Healthcare projects are some of the most complicated kinds of design projects. They have many different clients or stakeholders, often with specific requirements [3]. The functions of the space frequently change with new developments [6]. Compared with other projects, there is a high amount of Fixtures, Furniture and Equipment (FF&E) that need to be allowed for. Much of this equipment is highly specialised and often features particular spatial and connection requirements that may not be immediately obvious to the design team. Previous research has indicated that stakeholders have difficulties understanding what these items are in 2D plans [4].

AEC professionals produce 3D models using BIM software. These BIM models are used to generate plans, sections and elevations typically seen in architectural design packages. However, they have the potential to do much more. Each element in a BIM model has the ability to store a wide range of metadata, including but not limited to; quantities, material properties, dimensions, lead times, thermal data, manufacturer, cost etc. While most BIM objects come with a limited amount of metadata pre-programmed in, designers have the ability to extend the dataset as required, providing the ability to add additional information to aid in the design review process [7]. It is hypothesised that by allowing the stakeholders access to some of this information they will be able to better understand and assess the design in an architectural review.

3 STUDY DESIGN

To test whether BIM metadata can be viewed effectively within a VR headset an Operating Unit of a hospital was designed and built in *Autodesk Revit* and brought into the *Unity Game Engine*. The Operating Unit was broken up into two unique modules; 1) Theatre Module and 2) Recovery Module. The *Health Planning Unit - Operating Unit* from the AusHFG [8] was used as a guiding document. Participants reviewed these two module designs in either VR or using PDF 2D drawings alongside a 3D model.

3.1 Prototype Development

The two modules were built in *Autodesk Revit*, a leading AEC BIM software. The AusHFG website provides *Revit* models of the standard hospital rooms including all required furniture, fixtures and equipment. These were used as a base and were re-textured, partially remodeled to look more realistic and additional metadata was added to each object, specifically a new *Review Comment* field and a *User Comment* field. Building this in *Revit* enabled plans, elevations and schedules typical to a traditional design review to be created. The Operating Unit modules were then exported using *Unity Reflect* into the *Unity Game Engine*. *Unity Reflect* is an AEC specific plugin that creates a live link between the *Revit* file and a *Unity* file. *Reflect* maintains the fidelity of the BIM metadata as well, enabling its use within the *Unity Game Engine*. This meant that the same models with identical metadata could be used for the VR condition and the PDF condition of the user study.

3.2 User Study

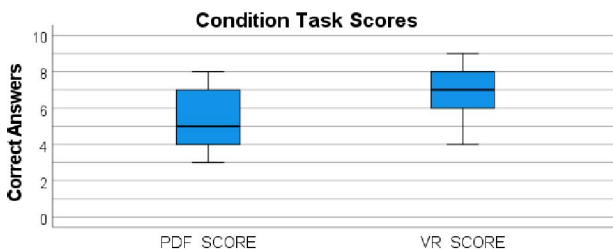
A within-subjects Latin square experiment design was used. Each participant reviewed one module in VR (see Figure 1) and the other module using PDF 2D drawings and a 3D model. The order in which participants used VR and PDF was alternated to allow for any possible experiment fatigue. A total of 17 healthcare professionals were recruited to conduct the user study. The roles in the field were varied with nurses, doctors, physiotherapists and pharmacists participating. To begin with, participants would review one module design in one condition, complete the tasks then answer an SUS questionnaire about that condition. They would then review the next module in the other condition, complete those tasks and answer an SUS questionnaire about that condition. Finally, a brief interview was conducted to talk directly about what they did / did not like about each condition.



Figure 1: Screen capture from user study.

The tasks the participants had to complete were designed to test their understanding of the design and their ability to provide feedback. The final task for each condition was to list any codewords found during the experiment. Within each module in each condition there were children's toys placed in unusual locations with codewords in their metadata. This was designed to test whether participants would notice obvious errors. Two participants in the VR condition found them during their experiment, one PDF user went back after being asked for the codewords but this was discounted as the purpose was to find them while completing the set tasks.

4 RESULTS



Preliminary results indicate that the VR condition was more successful with more participants getting more correct answers. A Wilcoxon signed-rank Test indicated that participants scored statistically significantly higher in the VR condition. $Z = -2.334$, $p = .020$. In the VR condition of the Theatre Module, participants averaged 6.5 correct answers (SD = 0.93), while the PDF condition of the Theatre Module participants averaged 4.22 correct answers (SD = 1.72). The VR condition of the Recovery module averaged 7.44 (SD = 1.51) correct answers while the PDF condition averaged 6.5 (SD = 1.20). Figure 2 shows the combined scores of the 2 different modules, i.e. the combined VR score had a mean of 7.00 (SD = 1.32), while the combined PDF score was 5.29 (SD = 1.86).

The VR condition also scored considerably higher on the SUS survey with 78.70 average SUS score (SD = 10.65) compared to 38.67 (SD = 19.31) for PDF and in the interview 16 of the 17 participants selected VR as their preferred design review condition.

5 DISCUSSION

The results show a clear preference for VR over the PDF condition. During the interview participants noted that it was much easier to find items in VR. They also mentioned that VR was more user friendly and intuitive. Some participants felt that the PDF drawings appeared to be more accurate and contain additional data. This is interesting as the VR condition and the PDF condition contained identical data sets. The participants generally stuck to the structured tasks and did not explore the modules further.

6 CONCLUSION AND FUTURE WORK

Overall, the participants found the VR condition easier to use and answered more questions correctly. However, few participants noticed the children's toys and were able to provide the hidden codewords. It is possible the participants were so focused on the tasks that they missed the obvious out of place objects. One way to test this would be to design an experiment that instead of guiding the participants to complete a series of structured tasks they could be placed within a room and asked to identify the errors themselves. With this user study design, the participants identified a clear preference and the high completion rate of the tasks point to VR being the superior review tool.

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